

About Simulation Purpose & Scope Standard - (SPSS)

Governing Why a Simulation Exists Before Deciding How It Should Represent Reality

Simulation Exists Before Deciding How It Should Represent Reality

A simulation may be technically sophisticated and still be unsuitable for the question it is being used to answer.

Before models, environments, assumptions, scenarios, inputs, or execution methods are selected, the simulation must establish why it is being performed and what it is legitimately intended to cover.

Simulation Purpose & Scope Standard (SPSS) provides this governance layer within SIMULOS®.

Canonical Definition

Simulation Purpose & Scope Standard (SPSS) defines the governance conditions under which the reason for performing a simulation, the questions it is intended to address, its governed scope, material inclusions and exclusions, applicability context, required simulation depth, and appropriate simulation precision are formally established before downstream simulation representation begins.

Why SPSS Exists

Organizations use simulations for different purposes: system design, engineering analysis, safety assessment, operational planning, autonomous-system testing, AI evaluation, infrastructure planning, scenario exploration, scientific investigation, training, decision support, and pre-deployment testing.

These purposes are not interchangeable.

A simulation designed to explore possible behavior may not provide sufficient basis for a safety-critical decision.

A simulation covering one operating environment cannot automatically support conclusions about another.

A simulation examining one configuration cannot silently become evidence for every configuration of the system.

SPSS prevents this purpose and scope inflation.

Governed Space

Simulation Purpose & Scope

SPSS governs the transition from:

Defined Simulation Object

to:

Governed Simulation Inquiry

It establishes what the simulation is expected to investigate before downstream simulation architecture is constructed around that objective.

Governance Flow

Simulation Object → Simulation Purpose → Simulation Question → Simulation Scope → Inclusions & Exclusions → Applicability Context → Simulation Depth → Simulation Precision → Governed Simulation Purpose & Scope

This governed basis then becomes an input to the next SIMULOS® space:

Reality Representation

What SPSS Establishes

A governed simulation should establish:

Purpose — why the simulation is being performed.

Simulation Question — what the simulation is expected to investigate or demonstrate.

Scope — which parts of the Simulation Object, system, process, environment, behavior, or interaction are included.

Exclusions — what the simulation explicitly does not represent or evaluate.

Applicability — the environments, configurations, populations, operating conditions, or contexts to which the simulation relates.

Simulation Depth — how extensively the simulation must investigate the defined problem.

Simulation Precision — what level of representational and analytical precision is appropriate for the intended purpose.

Business Value

For organizations using simulation, SPSS creates a clear governance basis before significant resources are committed to simulation design and execution.

It helps organizations determine whether: the simulation being built actually

addresses the business or engineering question, simulation expenditure is aligned with the intended decision, critical operating conditions have been excluded, simulation conclusions are being used beyond their legitimate scope, different simulation projects are addressing comparable questions, and downstream teams understand exactly what the simulation is and is not intended to establish.

This becomes increasingly important when simulation results influence investment, engineering, deployment, safety, regulatory, autonomous-system, or high-consequence operational decisions.

Governance Boundary

SPSS begins when a sufficiently identified Simulation Object requires a defined simulation purpose and governed operational scope.

It ends when the simulation purpose, questions, scope, exclusions, applicability context, required depth, and appropriate precision have been sufficiently established to support governed Reality Representation.

SPSS does not determine whether the simulation is valid, whether its evidence is sufficient for external validation, or whether operational reliance should ultimately be authorized.

Those questions belong to their appropriate downstream SIMULOS® spaces or complementary OOF® architectures.

Practical Governance Test

Before simulation development proceeds, an organization should be able to answer:

What exactly are we trying to learn, test, examine, or demonstrate through this simulation?

Which reality does that question concern?

What is inside the simulation scope?

What is explicitly outside it?

Under which conditions will the resulting simulation remain applicable?

How much simulation depth and precision does the intended purpose actually require?

If these questions cannot be answered, downstream simulation sophistication cannot compensate for the missing governance foundation.

Relationship to SIMULOS®

SPSS is the second Parent Standard of SIMULOS® — Simulation Governance Architecture.

It follows:

Simulation Object Standard

and precedes:

Reality Representation governance.

Its architectural role is fundamental:

The Simulation Object establishes what is being simulated.

SPSS establishes why it is being simulated and what the simulation is allowed to cover.

Only after both are established should the architecture begin governing how relevant reality will be represented.

Foundational Principle

A simulation cannot legitimately answer a broader question than the purpose and scope for which it was actually designed.

Canonical Closing Statement

Simulation Purpose & Scope Standard establishes the governed inquiry boundary of SIMULOS® by defining why a simulation is performed, what questions it is intended to address, what it includes and excludes, where its applicability begins and ends, and what level of depth and precision its purpose requires. By establishing these conditions before reality representation, model construction, scenario design, and execution, SPSS prevents simulation results from acquiring broader meaning, authority, or applicability than the simulation was actually designed to support.

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